

EXPERIMENT-3

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Comparative Analysis of GRU and Transformer Models for Time-Series Classification :-

1. Introduction :-

This project focuses on the classification of multivariate time-series sensor data using deep learning techniques. Two major architectures were implemented and analyzed:

- Recurrent Neural Network (GRU-based)
- Transformer-based model using self-attention

The goal was to evaluate their effectiveness in learning temporal patterns and improving classification accuracy.

2. Dataset and Preprocessing :-

The dataset consists of sequential sensor readings (e.g., accelerometer and gyroscope data) structured into fixed-length sequences.

Preprocessing Steps:

- Normalization of input features
- Feature engineering (magnitude, jerk, statistical features)
- Label encoding (converted to integer classes 0–4)
- Data split into training, validation, and testing sets

3. Model Architectures :-

3.1 GRU Model

The GRU model processes sequences step-by-step, maintaining temporal dependencies using gated mechanisms.

Advantages:

- Strong sequential understanding
- Stable training
- Works well with moderate datasets

3.2 Transformer Model

The Transformer uses self-attention to process all time steps simultaneously.

Key Components:

- Positional Encoding
- Multi-Head Attention
- Feed Forward Networks
- Layer Normalization

Advantages:

- Captures global dependencies
- Parallel computation
- Better feature interaction learning

4. Training Details :-

GRU Model:

- Fewer epochs used initially
- Sequential processing
- Higher time per epoch

Transformer Model:

- Trained for more epochs (up to 50)
- Learning rate scheduling applied (ReduceLROnPlateau)
- Faster computation per epoch due to parallel processing

5. Performance Results :-

Transformer Final Results (from logs):

- Training Accuracy: ~90.5%
- Validation Accuracy: ~88.3%
- Validation Loss: ~0.29

GRU Results:

- Training Accuracy: ~86–88%
- Validation Accuracy: ~85–87%

6. Key Observations :-

6.1 Accuracy Comparison

The Transformer achieved higher accuracy compared to GRU, with an improvement of approximately 2–3% in validation accuracy.

6.2 Effect of Epochs

- The Transformer was trained for more epochs, allowing it to learn richer representations.
- GRU performance was slightly lower partly because:
 - It was trained for fewer epochs
 - It converges slower due to sequential processing

Important Insight:

If GRU were trained for more epochs, it could potentially achieve similar accuracy.

6.3 Training Time Comparison :-

Model	Time per Epoch	Training Style
GRU	Higher ⌚	Sequential
Transformer	Lower ⚡	Parallel

- GRU processes data step-by-step → slower
- Transformer processes entire sequence at once → faster

6.4 Learning Behaviour :-

GRU:

- Learns temporal dependencies naturally
- Stable and less sensitive to hyperparameters

Transformer:

- Learns global relationships via attention
- Requires tuning but gives better results

6.5 Generalization :-

- Transformer shows strong generalization with small gap between training and validation accuracy
- Learning rate scheduling helped prevent overfitting

7. Confusion Matrix Analysis :-

- Most predictions lie along the diagonal → high accuracy
- Errors occur between similar classes
- Transformer shows slightly better class separation than GRU

8. Why Transformer Outperformed GRU :-

The Transformer achieved better performance due to:

- Self-attention capturing long-range dependencies
- Parallel processing improving learning efficiency
- Increased training epochs
- Learning rate optimization
- Better feature interaction modeling

9. Important Comparative Insight :-

While Transformers achieved higher accuracy, GRU models are still highly effective for time-series data. With sufficient training (more epochs), GRU can achieve comparable performance. However, GRU requires more computation time per epoch due to its sequential nature, whereas Transformers are computationally efficient due to parallelism.

10. Conclusion :-

This study demonstrates that:

- Transformer models outperform GRU in terms of accuracy and efficiency when properly tuned
- GRU remains a strong baseline model for sequential data
- Model selection depends on trade-offs between accuracy, computation time, and data size

11. Final Statement :-

The Transformer model achieved superior performance due to its ability to capture global dependencies and train efficiently. However, GRU models remain competitive and can achieve similar accuracy with extended training, though at the cost of higher computational time per epoch.

Graphs :-

